

Solar Storage, Statistics, and Cement; Picking Batteries Has Never Been Easier

Why Solar?

With the issue of climate change rapidly worsening and costs of living soaring higher than ever, more and more people are looking to go off the grid and become self-sufficient. Without an electricity company constantly pumping energy to their home, homeowners must utilize solar panels to support their daily life.

Energy Storage

Of course, it is not sunny *all* of the time, which sprouts a risk in this constant reliance on the sun. Just because the sun has gone down does not mean that homeowner energy needs have ceased. This is where the importance of an effective energy storage system becomes prevalent. The best storage system must store enough energy to power the home on days in which energy demand is high but also cost the homeowner as little as possible.

User-Friendly Decision Making

To decide upon the optimal number and brand of batteries to power a home, the homeowner needs to do very little work. When using the model, they simply input key details about their daily schedule and energy usage. Even the kilowattage of each appliance can be edited to customize the decision.

Kilowatt Hour Calculation

From the provided information, trends are interpreted on which hours the home will be using the most energy. This helps establish which hours rely solely on continuous power and which will also require instantaneous power.

The Final Choice

In congruence with the energy demand by the hour, climate data is analyzed to see when the home will be relying on its battery storage system. This finalizes the exact energy storage requirements and then matches it with the best batteries for the output requirements and sunlight availability.

Cement Batteries

There is an even newer form of energy storage on the horizon - cement batteries. These batteries have a variety of benefits that make them appealing to almost every homeowner, predominately the fact that they are already integrated into the very foundation of many American homes. With a little bit more analysis and simple integration into the code, the model will soon be able to analyze and compare current batteries on the market with these new cement batteries.